Krishnaa **Vadivel** | Embedded Systems Engineer | Firmware | FPGA | Robotics

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Summary

Embedded systems engineer with hands-on experience across MCU firmware, FPGA logic, sensor-logging systems, robotics, and engineering validation. Comfortable moving from low-level C and digital logic through oscilloscope-based debugging, field testing, and host-side tooling for visualization and diagnostics.

Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer — *2019–2021*

- Developed embedded sensor-logging and control functionality for geological well tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class platforms.
- Supported board- and system-level bring-up with oscilloscope-based debugging, field validation, and targeted PCB updates in Altium.
- Built companion CUDA/OpenGL engineering tools for waveform processing, visualization, and diagnostics around the embedded hardware stack.

Yale University
Embedded Systems Teaching Assistant / PhD Researcher — *2021–present*

- Supported embedded systems instruction and student lab work involving firmware, hardware interfaces, debugging, and instrumentation.
- Brought a strong embedded perspective into broader scientific-computing and engineering workflows during PhD research.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026, strengthening the research side of my hardware-software profile.

Selected Projects

EMG Robotic Hand: Embedded systems project combining surface EMG sensing, analog filtering, PIC32 control logic, and servo actuation; related work was published in Circuit Cellar.

Maze Mapping Robot: Course-based robotics work involving navigation logic, wireless coordination, and remote-station control.

MIPS Processor: Single-cycle Verilog CPU with self-checking test bench for readable digital-design validation.

Matrix-Multiply Accelerator: Small Verilog accelerator block showing datapath and control concepts relevant to embedded ML hardware.

Education

Yale University
PhD, Electrical Engineering — *2021–present*

Yale University
MPhil, Electrical Engineering — *2023*

Yale University
MS, Electrical Engineering — *2023*

Cornell University
MEng, Electrical and Computer Engineering — *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Firmware: C, C++, MCU firmware, hardware-software integration
Hardware: Verilog, FPGA, PIC32, dsPIC, STM32, C8051, sensor interfaces
Debug: Oscilloscopes, bench instrumentation, field testing, board bring-up
Software: Python, Linux, CUDA, OpenGL
Interfaces: SPI, I2C, UART, embedded control and logging workflows